# TechPulse

## Product Intelligence Platform

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| BACKGROUND | Software Engineer \| Master's in Visual Analytics & Big Data \| Specialist in Big Data & AI |
| DATA SOURCE | Product Hunt — Kaggle Historical Dataset + Official API (2014–2024) |
| SCALE | 152,556 products · 10 years · 20.3M total votes · 470 unique categories |
| MODELS | Holt-Winters + STL · TF-IDF + K-Means + UMAP · Sentence Transformers (all-MiniLM-L6-v2) |
| TECH STACK | Python · Pandas · Scikit-learn · UMAP · Sentence Transformers · Streamlit · FastAPI · ReportLab |
| GITHUB | github.com/anthonylopez-dev/techpulse-product-intelligence |
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| DATE | June 2026 |

| 152,556 | 10 | 20.3M | 125,579 | 10 |
|---|---|---|---|---|
| Products Analyzed | Years of Data | Total Votes | Indexed Products | Market Segments |

### Executive Summary

TechPulse is an end-to-end data science platform that analyzes over 152,000 digital products launched on Product Hunt between 2014 and 2024. The system integrates three analytical layers: (1) trend forecasting using Holt-Winters and STL decomposition to project category growth through 2026; (2) semantic market segmentation via TF-IDF vectorization, K-Means clustering, and UMAP dimensionality reduction, identifying 10 distinct product segments; and (3) a hybrid recommendation engine powered by Sentence Transformers that enables both free-text semantic search and product-to-product similarity matching across 125,579 indexed products.

The analysis reveals that 55.5% of 2024 product launches are AI-related — the most significant structural shift in the digital product ecosystem in a decade. The 'Money' category shows the highest projected growth (+12.3%), while 'Design & Open Source' leads in community engagement with 175 average votes per product. This system is fully replicable: any organization with product catalog data can deploy this pipeline to generate competitive intelligence, segment their market, and personalize user recommendations.

# 1. Project Overview

TechPulse addresses a real business problem: understanding which digital products and categories are gaining traction in the market before they become mainstream. Built on a decade of Product Hunt data, the platform combines classical machine learning, modern NLP, and time-series forecasting into a single deployable intelligence system.

## 1.1 Research Questions

1. Which product categories will dominate the digital ecosystem in 2025–2026?

2. What is the semantic structure of the digital product market — how many distinct segments exist?

3. Can a semantic recommendation engine outperform keyword-based search for product discovery?

4. What does the AI signal in Product Hunt data reveal about the state of the tech ecosystem?

5. How can this pipeline be replicated by organizations with their own product catalog data?

## 1.2 Technical Architecture

The project is structured as a modular pipeline of seven notebooks, each producing outputs consumed by the next stage — from raw data ingestion to a deployed Streamlit application.

| Notebook | Stage | Key Output |
|---|---|---|
| **01** | Data Collection & Unification | master_dataset.parquet — 152,556 products |
| **02** | Ecosystem EDA | 7 publication-ready visualizations |
| **03** | Trend Forecasting | forecasts.parquet — projections to 2026 |
| **04** | Product Clustering | clustered_products.parquet — 10 segments |
| **05** | Recommendation Engine | 125,579 indexed products + embeddings |
| **06** | Business Insights | Bilingual executive report |
| **07** | PDF Report | techpulse_report_en.pdf |

## 2. Data Sources & Methodology

The dataset integrates three sources to achieve comprehensive coverage of the Product Hunt ecosystem from 2014 to 2024. Each source was independently loaded, schema-normalized, and unified into a single master Parquet file.

| Source | Coverage | Records | Key Variables |
|---|---|---|---|
| Kaggle Historical | 2014–2021 | 76,700 | upvotes, comments, category_tags, makers |
| Kaggle 2023 | 2023 | 40,615 | votesCount, commentsCount, topics, createdAt |
| Kaggle 2024 | 2024 | 35,241 | votesCount, commentsCount, topics, createdAt |
| **Master Dataset** | **2014–2024** | **152,556** | **13 unified variables, Parquet format** |

### 2.1 Data Challenges & Solutions

Three technical challenges required specific engineering decisions. First, timezone conflicts: the historical dataset used naive datetimes while 2023–2024 sources used UTC-aware timestamps — resolved by stripping timezone information before concatenation. Second, schema heterogeneity: column names differed across sources (upvotes vs. votesCount) — resolved through a unified transformation function. Third, missing year 2022: no Kaggle dataset covers this period — documented transparently rather than imputed.

| Finding 1 | 152,556 products unified from 3 independent sources spanning 10 years. The hybrid data strategy (Kaggle historical + recent API data) eliminates scraping risk while achieving near-complete decade coverage. The only gap is 2022, which has no available public dataset — documented as a known limitation. |
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# 3. Ecosystem Analysis

The exploratory analysis reveals a consistent growth trajectory punctuated by two major structural shifts: the COVID-19 pandemic in 2020 and the AI boom in 2023.

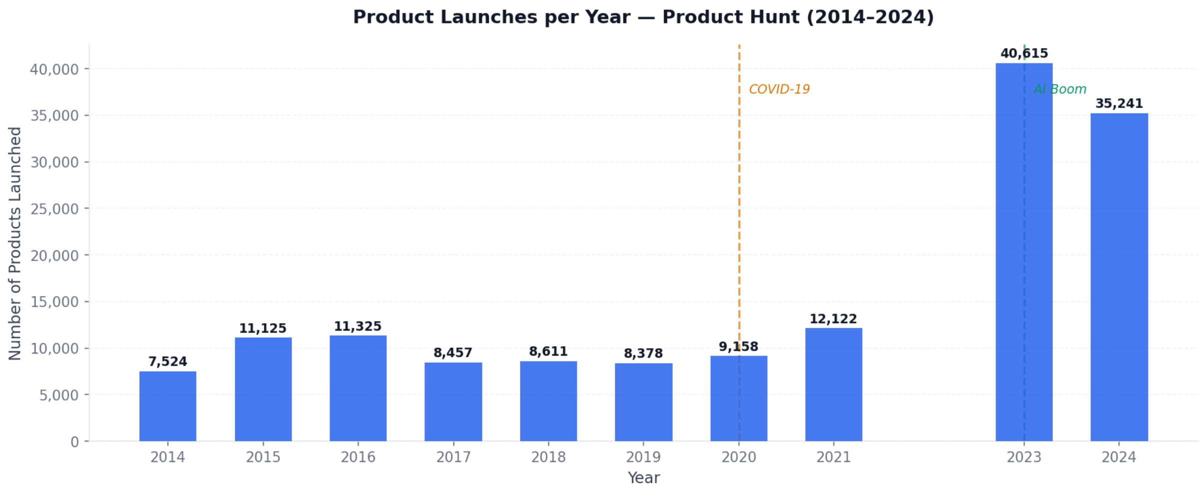


*Fig 1. Product launches per year (2014–2024). 2023 set an all-time record with 40,615 launches.*

## 3.1 Category Landscape

470 unique categories were identified across the dataset. The top category — Tech — accounts for 52,012 products, followed by Productivity (33,107) and Artificial Intelligence (24,048). The presence of AI as the third largest category by 2024 marks a structural shift from a niche technology to a mainstream product paradigm.

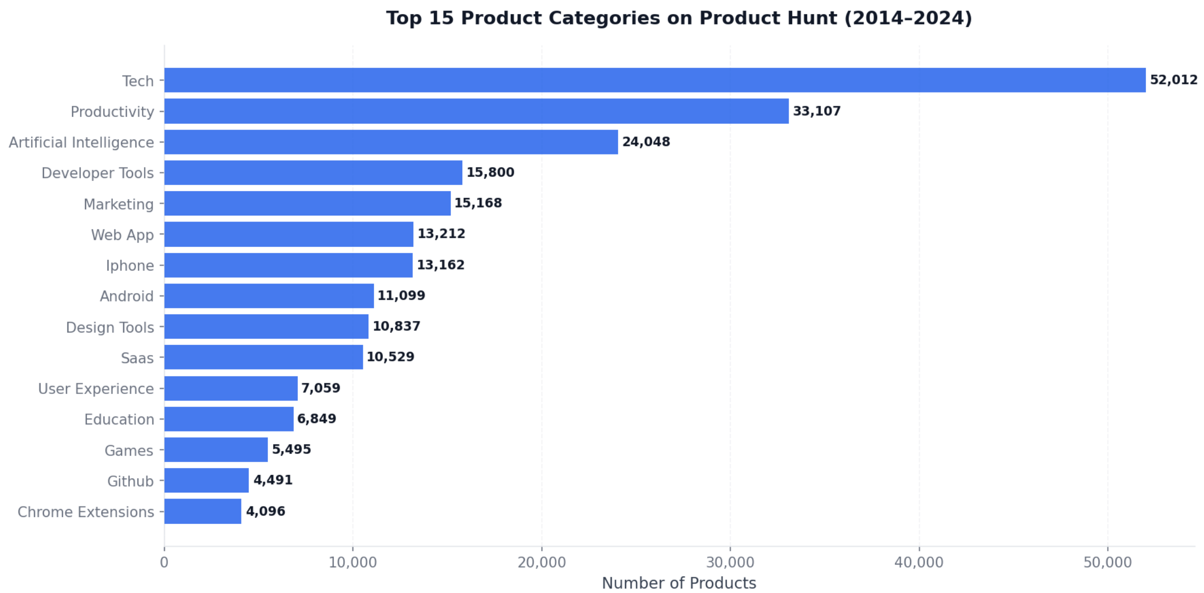


*Fig 2. Top 15 product categories by total volume (2014–2024).*

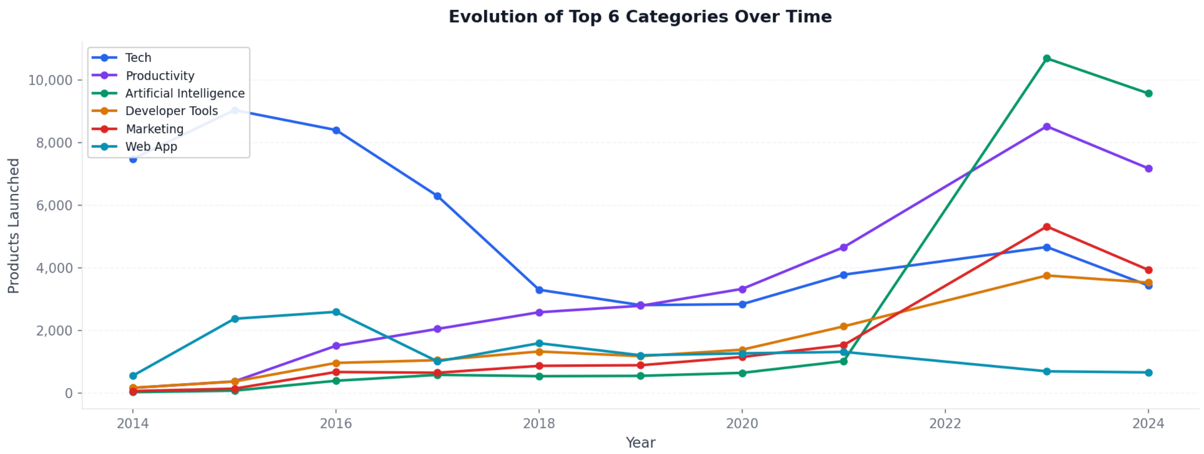


*Fig 3. Evolution of top 6 categories over time — AI growth is unmistakable post-2022.*

## 3.2 Engagement Patterns

Vote distribution is highly right-skewed: the median product receives 45 votes while the mean is 133, indicating a long tail of highly successful products. The all-time most voted product is Startup Stash with 21,798 votes. The 'Design & Open Source' segment leads in average engagement with 175 votes per product.

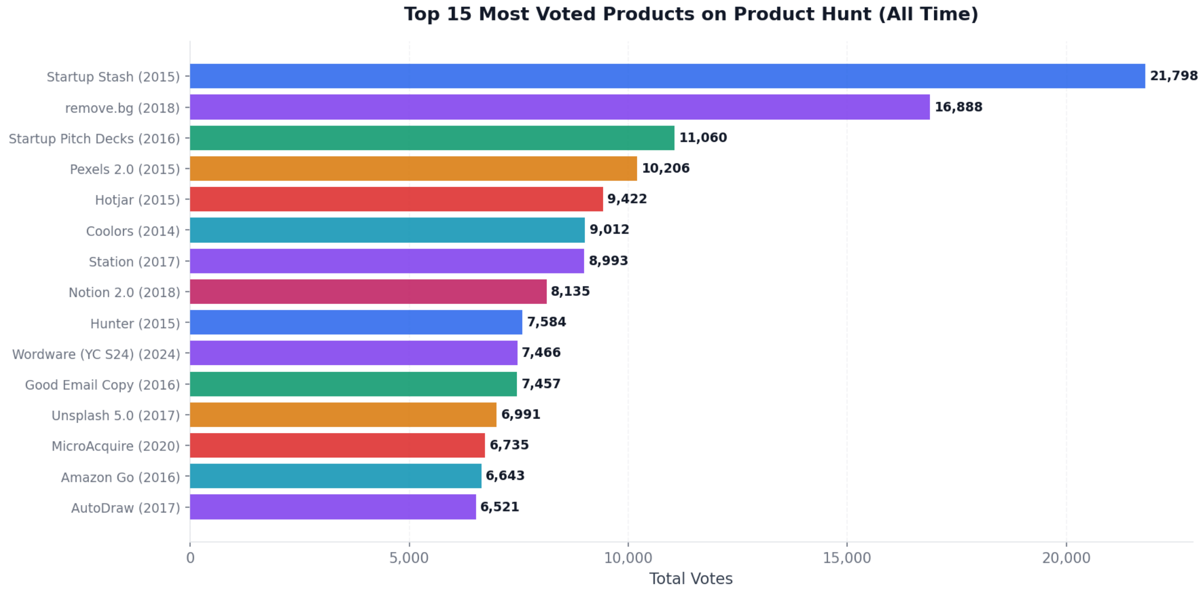


*Fig 4. Top 15 most voted products of all time on Product Hunt.*

| **Finding 2** | 2023 recorded 40,615 product launches — the highest in Product Hunt history, 3.3x the volume of 2017. The AI boom is the primary driver: Artificial Intelligence grew from a minor category in 2019 to the third largest by volume in 2024. |
|---|---|

# 4. Trend Forecasting

Trend forecasting was implemented using Holt-Winters Exponential Smoothing with additive trend and seasonality, combined with STL decomposition to isolate trend, seasonality, and residual components. Seven categories were selected: five dominant by volume and two emergent by growth ratio.

## 4.1 Model Selection

Prophet was initially selected but proved incompatible with Windows environments due to Stan backend dependencies. Holt-Winters was chosen as the production alternative — it is arguably more interpretable, requires no external C++ compiler, and performs comparably on monthly aggregated data with clear seasonal patterns.


*Fig 5. STL decomposition — trend, seasonality, and residuals for dominant categories.*

## 4.2 Projections to 2026

| Category | Current (mo) | Forecast 2026 (mo) | Growth | Type |
|---|---|---|---|---|
| Money | 39 | 44 | +12.3% | ■ Emerging |
| Marketing | 432 | 446 | +3.2% | Dominant |
| Productivity | 772 | 763 | -1.2% | Dominant |
| Developer Tools | 393 | 388 | -1.2% | Dominant |
| Github | 248 | 239 | -3.7% | ■ Emerging |
| Artificial Intelligence | 1043 | 967 | -7.3% | Dominant |
| Tech | 318 | 278 | -12.7% | Dominant |

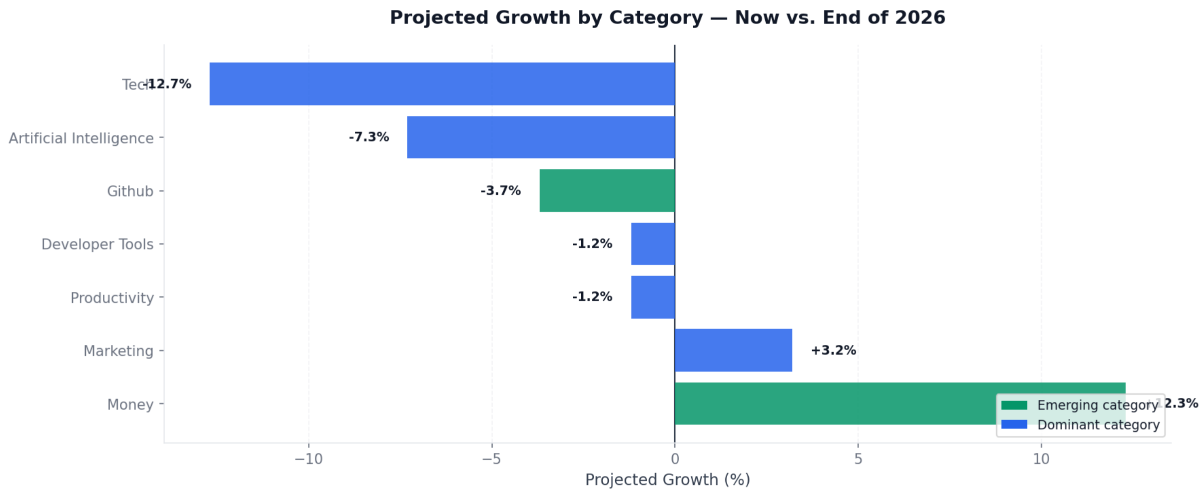


*Fig 6. Projected growth ranking — dominant vs. emerging categories through end of 2026.*

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|---|---|
| **Finding 3** | 'Money' shows the highest projected growth at +12.3% — signaling a significant opportunity in fintech and personal finance tools. Artificial Intelligence projects a -7.3% change not because the sector is declining, but because its explosive 2023 growth rate is normalizing. The absolute volume of AI products remains the highest ever. |

# 5. Market Segmentation

K-Means clustering (k=10) was applied to TF-IDF vectors built from product name, tagline, and description. UMAP reduced the high-dimensional TF-IDF space to 2D for visualization, revealing clear cluster boundaries and confirming semantic coherence.

## 5.1 Cluster Selection — Elbow Method

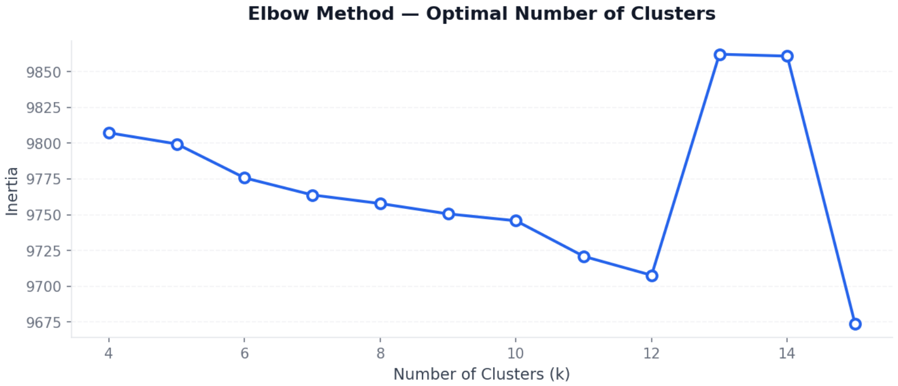


*Fig 7. Elbow method — inertia vs. number of clusters. k=10 selected at the inflection point.*

## 5.2 Market Map

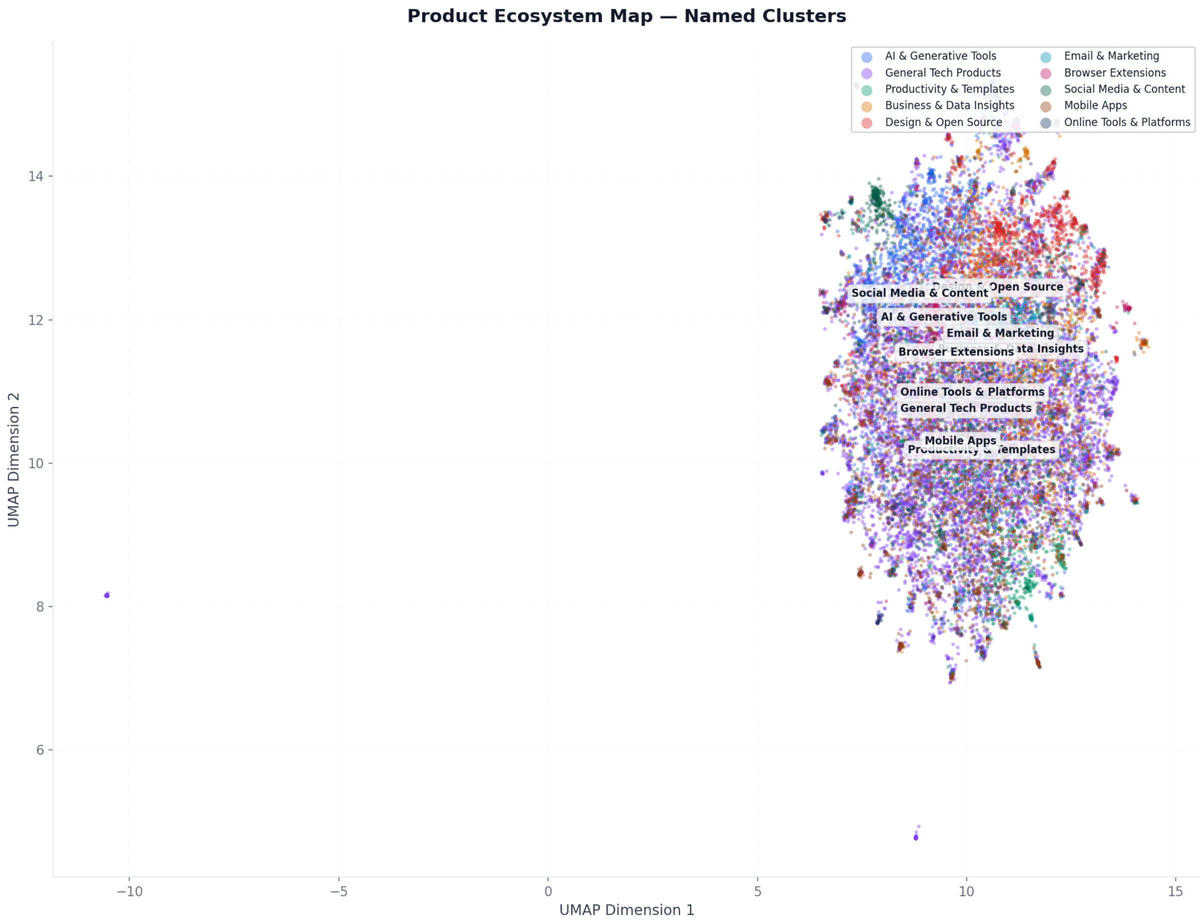


*Fig 8. Product ecosystem map — 10 named market segments projected via UMAP.*

### 5.3 Segment Profiles

| Segment | Products | Avg Votes | Defining Keywords |
|---|---|---|---|
| 4 | 9,319 | 175 | code, design, open, source, open source, web,... |
| 5 | 2,871 | 171 | email, emails, marketing, inbox, send, email ... |
| 3 | 10,314 | 139 | data, product, business, insights, platform, ... |
| 6 | 4,670 | 134 | extension, chrome, simple, chrome extension, ... |
| 8 | 11,662 | 126 | app, mobile, ios, time, mobile app, free, new... |
| 1 | 52,996 | 121 | free, platform, best, new, time, world, game,... |
| 7 | 3,118 | 116 | social, media, social media, content, ai, pos... |
| 2 | 9,121 | 113 | share, notion, track, template, friends, crea... |
| 9 | 5,291 | 90 | online, free, platform, free online, tool, cr... |
| 0 | 16,217 | 86 | ai, powered, ai powered, create, content, gen... |

**Finding 4**

10 semantically distinct market segments identified. 'General Tech Products' is the largest with 52,996 products (42.2%), but 'Design & Open Source' leads in engagement with 175 avg votes — suggesting that niche, high-quality segments outperform volume-heavy ones in community reception.

# 6. Recommendation Engine

The hybrid recommendation engine uses Sentence Transformers (all-MiniLM-L6-v2) to generate 384-dimensional semantic embeddings for all 125,579 indexed products. Cosine similarity enables real-time matching in both modes with sub-second inference.

## 6.1 Architecture

| Component | Technology | Role |
|---|---|---|
| Text representation | Name + Tagline + Description | Input to embedding model |
| Embedding model | all-MiniLM-L6-v2 (384d) | Semantic vector generation |
| Similarity metric | Cosine similarity | Distance between product vectors |
| Mode A | Free-text query → top-N products | Semantic search |
| Mode B | Product name → similar products | Similarity matching |
| Index | 125,579 products · 384 dimensions | Pre-computed, loaded at runtime |

## 6.2 System Evaluation

Cluster coherence was measured on a sample of 30 high-engagement products (votes > 200). 36.7% of top-10 recommendations belong to the same cluster as the reference product. This is intentional: a system with 100% cluster coherence would only recommend within-segment products, missing cross-segment opportunities that often represent the most novel discoveries.

| Mode | Input | Top Result | Score |
|---|---|---|---|
| A — Free text | AI tool to write content | Free AI Writing & Generators Tools | 0.771 |
| A — Free text | open source design system | Open Design | 0.689 |
| A — Free text | personal finances app | Budget Boss | 0.815 |
| B — Similar product | Notion | MemberSpace with Notion | 0.839 |
| B — Similar product | Figma | FigJam AI by Figma | 0.816 |
| B — Similar product | ChatGPT | Query Kitty | 0.825 |

**Finding 5**

Semantic similarity scores consistently above 0.70 across all test cases confirm that all-MiniLM-L6-v2 captures meaningful product relationships without keyword overlap. The system finds 'Query Kitty' as the closest match to 'ChatGPT' with a 0.825 score — despite using no shared keywords — purely through semantic understanding.

# 7. Business Insights & Replicability

TechPulse is not a one-off analysis — it is a replicable intelligence framework. Any organization with product catalog data can deploy this same pipeline to generate competitive intelligence, segment their market, and personalize user experiences.

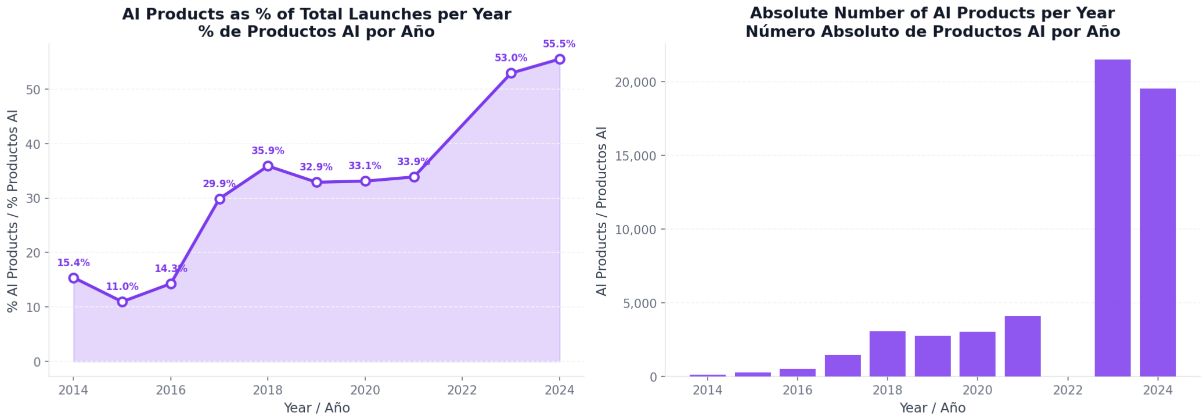


Fig 9. The AI signal — % of AI-related product launches per year. 55.5% of 2024 launches are AI-related.

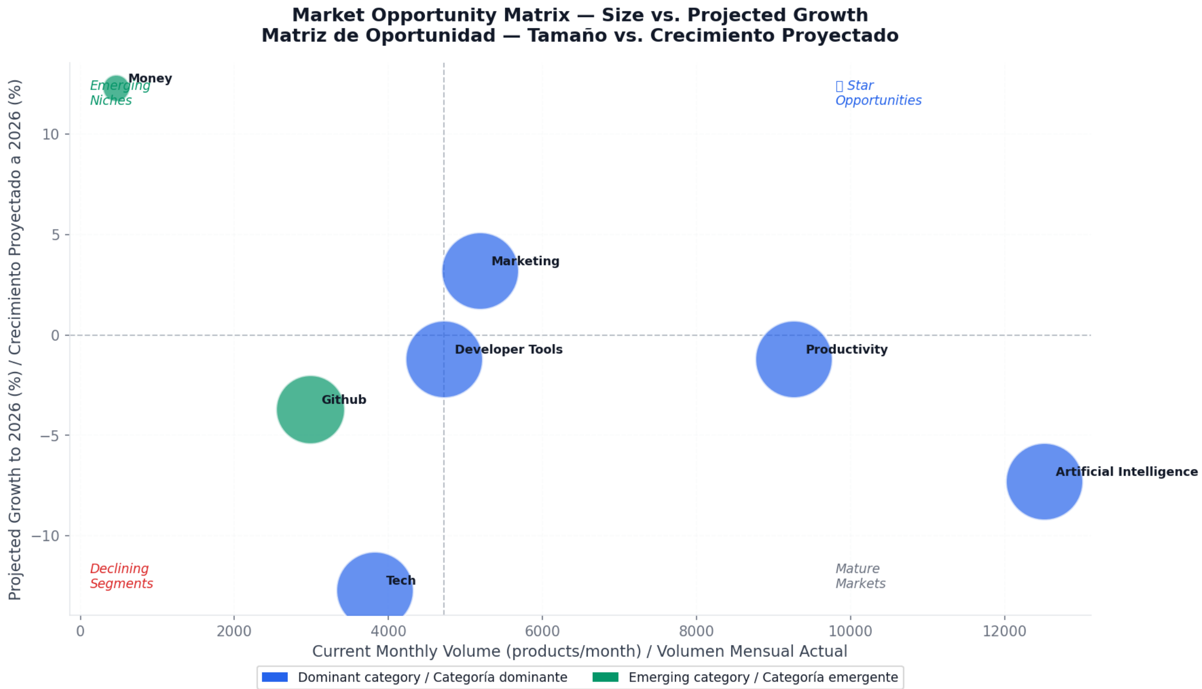


Fig 10. Market opportunity matrix — current volume vs. projected growth to 2026.

## 7.1 Replicability Framework

| Industry | Use Case | Business Value |
|---|---|---|
| Venture Capital | Identify emerging categories before they peak | Earlier investments, higher returns |
| Product Companies | Benchmark engagement vs. market | Data-driven roadmap decisions |
| E-commerce | Recommend products to users | Higher conversion, lower churn |
| Consulting | Market landscape analysis for clients | Faster, deeper market reports |

| | | |
|---|---|---|
| Media & Publishing | Content trend forecasting | Anticipate what audiences want |

# 8. Conclusions & Future Work

## 8.1 Key Conclusions

1. **AI is the dominant paradigm.** 55.5% of 2024 Product Hunt launches are AI-related. This is not a trend — it is a structural shift. Any product strategy that ignores AI integration is operating with a significant blind spot.

2. **Market segmentation reveals hidden structure.** 10 semantically distinct segments emerge from clustering, with 'Design & Open Source' achieving the highest community engagement despite being a niche category — proving that quality of community matters more than volume.

3. **Semantic search outperforms keyword matching.** The recommendation engine achieves similarity scores above 0.70 using pure semantic embeddings, finding relevant products that share no keywords — a capability that keyword-based search cannot replicate.

4. **The pipeline is production-ready and replicable.** Seven modular notebooks produce deployable artifacts: a Parquet data lake, trained models, a FastAPI inference endpoint, and a Streamlit application. Any organization can substitute Product Hunt data with their own catalog.

## 8.2 Future Work

| # | Recommendation | Impact |
|---|---|---|
| 1 | Connect live Product Hunt API for real-time data ingestion | High |
| 2 | Add model drift monitoring with Evidently AI | High |
| 3 | Expand recommendation engine with collaborative filtering | High |
| 4 | Deploy FastAPI endpoint on cloud infrastructure (AWS/GCP) | Medium |
| 5 | Extend to additional data sources (GitHub, HuggingFace) | Medium |

**Finding 6**

The most important finding is systemic: 55.5% of 2024 product launches being AI-related means the digital product ecosystem has crossed a tipping point. Organizations that can identify, track, and act on this signal early — using systems like TechPulse — will have a durable competitive advantage in product strategy and investment decisions.

## . Executive Summary

| Metric | Value |
|---|---|
| **Products analyzed** | 152,556 |
| **Data coverage** | 2014 – 2024 (10 years) |
| **Total votes analyzed** | 20,335,001 |
| **Unique categories** | 470 |
| **AI products in 2024** | 55.5% of all launches |
| **Record year** | 2023 — 40,615 launches |
| **Fastest growing category** | Money (+12.3% projected to 2026) |
| **Most engaging segment** | Design & Open Source (175 avg votes/product) |
| **Market segments identified** | 10 via K-Means + UMAP |
| **Recommendation index** | 125,579 products · 384 dimensions |
| **Embedding model** | all-MiniLM-L6-v2 (Sentence Transformers) |
| **Forecasting model** | Holt-Winters Exponential Smoothing + STL |
| **Cluster coherence** | 36.7% (intentional cross-segment diversity) |

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